

Cosets, Lagrange's Theorem, and groups of small order

Sections 5.2 and 5.3 of *Numbers, Groups and Codes*

Applied Algebra
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Block	Main focus
1	Cosets: definition, examples, partitions, and the equal-size theorem.
2	Lagrange's Theorem: proof and consequences for element orders, prime-order groups, Fermat, and Euler.
3	Groups of small order: isomorphism, direct products, and the classification up to order 8.

Cosets

Translating a subgroup around a group

Recall: subgroups and cyclic subgroups

Let G be a group.

- A non-empty subset $H \subseteq G$ is a **subgroup**, written $H \leq G$, if it is a group under the same operation.
- Subgroup test:

$$H \leq G \iff h, k \in H \Rightarrow hk^{-1} \in H.$$

- The cyclic subgroup generated by g is

$$\langle g \rangle = \{g^n : n \in \mathbb{Z}\}.$$

- If $\text{ord}(g) = n$, then

$$\langle g \rangle = \{e, g, g^2, \dots, g^{n-1}\}.$$

Why introduce cosets?

A subgroup H is a smaller copy of group structure inside G .

A coset is a **translate** of this subgroup.



This idea gives a powerful counting method: split G into equal-sized pieces.

Definition: left and right cosets

Let $H \leq G$ and let $a \in G$.

The **left coset** of H represented by a is

$$aH = \{ah : h \in H\}.$$

The **right coset** of H represented by a is

$$Ha = \{ha : h \in H\}.$$

Convention in this lecture:

When we say “coset” without qualification, we mean a left coset.

Basic facts about cosets

Let $H \leq G$ and $a \in G$.

- $H = eH$, so H is a coset of itself.
- $a \in aH$, since $a = ae$.
- If $b \in aH$, then $bH = aH$.
- Thus a coset is not represented by one unique element: every element in the coset is a representative.
- Usually aH is **not** a subgroup unless $a \in H$.

Two trivial cases:

$$H = G \Rightarrow \text{one coset}, \quad H = \{e\} \Rightarrow \text{singleton cosets.}$$

Example: congruence classes are cosets

Take the additive group $(\mathbb{Z}, +)$.

Let

$$H = n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}, \quad n \geq 2.$$

Then $H \leq \mathbb{Z}$, and the coset of k is

$$k + H = \{k + nh : h \in \mathbb{Z}\}.$$

But this is exactly the congruence class of k modulo n :

$$k + n\mathbb{Z} = [k]_n.$$

So cosets generalize the familiar partition of $\mathbb{Z}$ into congruence classes.

Example: cosets in $\mathbb{Z}_6$

Let

$$G = (\mathbb{Z}_6, +), \quad H = \{[0]_6, [3]_6\} = \langle [3]_6 \rangle.$$

For readability, write $0, 1, \dots, 5$ for the congruence classes.

The cosets are

$$0 + H = \{0, 3\} = 3 + H,$$

$$1 + H = \{1, 4\} = 4 + H,$$

$$2 + H = \{2, 5\} = 5 + H.$$

Each coset has 2 elements, the same number as H .

Example: cosets in S_3

Let

$$G = S_3, \quad H = \{\text{id}, (1\ 2)\}.$$

The left cosets are

$$H = \{\text{id}, (1\ 2)\},$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\},$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\}.$$

There are three cosets, each with two elements.

This example also shows why non-Abelian groups require care.

Left cosets and right cosets need not agree

In an Abelian group,

$$aH = Ha$$

because $ah = ha$ for all $h \in H$.

In a non-Abelian group, the two can differ.

Using the previous S_3 example,

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\},$$

while

$$H(1\ 3) = \{(1\ 3), (1\ 3\ 2)\}.$$

So the list of left cosets and the list of right cosets are not necessarily the same.

Theorem 5.2.1: cosets are equal or disjoint

Theorem. Let $H \leq G$ and let $a, b \in G$. Then

$$aH = bH \quad \text{or} \quad aH \cap bH = \emptyset.$$

Proof idea. Suppose $aH \cap bH \neq \emptyset$. Choose

$$c \in aH \cap bH.$$

Since $c \in aH$, the representative fact gives

$$cH = aH.$$

Since $c \in bH$, the same fact gives

$$cH = bH.$$

Therefore $aH = bH$.

Equivalence relation viewpoint

There is another useful way to see the partition.

Define a relation on G by

$$x \sim y \iff x^{-1}y \in H.$$

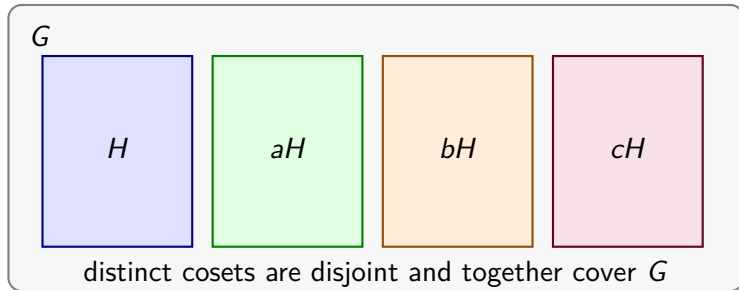
Then $\sim$ is an equivalence relation:

- reflexive: $x^{-1}x = e \in H$;
- symmetric: if $x^{-1}y \in H$, then $(x^{-1}y)^{-1} = y^{-1}x \in H$;
- transitive: if $x^{-1}y \in H$ and $y^{-1}z \in H$, then

$$x^{-1}z = (x^{-1}y)(y^{-1}z) \in H.$$

The equivalence class of x is exactly xH .

Cosets partition the group



Every element of G belongs to exactly one left coset of H .

Theorem 5.2.2: all cosets have the same size

Theorem. Let $H \leq G$. Each coset aH has the same number of elements as H .

Bijection. Define

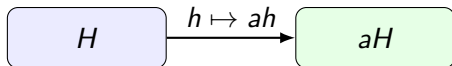
$$f : H \longrightarrow aH, \quad f(h) = ah.$$

Injective:

$$ah_1 = ah_2 \Rightarrow h_1 = h_2$$

by cancellation.

Surjective: Every element of aH has the form ah .



Theorem 5.2.3: Lagrange's Theorem

Notation. The number of elements of a group G is denoted by $o(G)$.

Let G be a finite group and let $H \leq G$.

Suppose H has m distinct left cosets in G .

Then

$$o(G) = m o(H).$$

Consequently,

$$o(H) \mid o(G).$$

The integer $[G : H] = m$ is called the **index** of H in G .

Proof of Lagrange's Theorem

List the distinct left cosets of H in G :

$$G = a_1H \cup a_2H \cup \cdots \cup a_mH, \quad a_1 = e.$$

By Theorem 5.2.1, this union is disjoint.

By Theorem 5.2.2, every coset has $o(H)$ elements.

Therefore

$$o(G) = \underbrace{o(H) + o(H) + \cdots + o(H)}_{m \text{ times}} = m o(H).$$

So $o(H)$ divides $o(G)$.

Interpretation and cautions

Lagrange's Theorem is a counting theorem:

$$\text{group size} = (\text{coset size}) \times (\text{number of cosets}).$$

Important cautions:

- A coset is usually not a subgroup.
- Left cosets and right cosets need not be the same sets.
- The number of left cosets equals the number of right cosets.
- The converse of Lagrange's Theorem is false in general:

$$d \mid o(G) \not\Rightarrow \text{there is a subgroup of order } d.$$

But Lagrange's Theorem gives strong restrictions on what can happen.

Consequence: element orders divide group order

Corollary 5.2.4. Let $g \in G$. Then

$$\text{ord}(g) \mid o(G).$$

Reason. The order of g is the size of the cyclic subgroup it generates:

$$\text{ord}(g) = o(\langle g \rangle).$$

Since $\langle g \rangle \leq G$, Lagrange's Theorem gives

$$o(\langle g \rangle) \mid o(G).$$

Example. If $o(G) = 20$, possible element orders are only

$$1, 2, 4, 5, 10, 20.$$

Consequence: groups of prime order are cyclic

Corollary 5.2.5. If G has prime order p , then G is cyclic.

Proof. Choose $x \in G$ with $x \neq e$.

Then $\langle x \rangle$ has more than one element.

By Lagrange's Theorem,

$$o(\langle x \rangle) \mid p.$$

Since p is prime and $o(\langle x \rangle) > 1$, we must have

$$o(\langle x \rangle) = p.$$

Thus $\langle x \rangle = G$.

Consequence: Fermat's Theorem

Let p be prime and let a be an integer not divisible by p .

Then $[a]_p$ lies in the multiplicative group G_p of invertible congruence classes modulo p .

This group has $p - 1$ elements:

$$o(G_p) = p - 1.$$

By Corollary 5.2.4, the order of $[a]_p$ divides $p - 1$.

Therefore

$$a^{p-1} \equiv 1 \pmod{p}.$$

This recovers Fermat's Theorem from group theory.

Consequence: Euler's Theorem

Let $n > 1$ and let a be relatively prime to n .

Then $[a]_n$ lies in the group G_n of invertible congruence classes modulo n .

The number of elements in G_n is Euler's phi-function:

$$o(G_n) = \varphi(n).$$

By Corollary 5.2.4,

$$\text{ord}([a]_n) \mid \varphi(n).$$

Therefore

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Quick checks

- 1 Let $G = (\mathbb{Z}_{12}, +)$ and $H = \{0, 4, 8\}$. List all cosets of H in G .
- 2 If G has order 18, what are the possible orders of an element of G ?
- 3 Explain why every group of order 11 is cyclic.
- 4 True or false: if $d \mid o(G)$, then G must have a subgroup of order d .

Groups of small order

Isomorphism, direct products, and classification

When are two groups “the same”?

Two groups may look different because their elements have different names.

They are **isomorphic** if their group structure is the same after relabelling.

Definition. An isomorphism from G to H is a bijection

$$\theta : G \longrightarrow H$$

such that

$$\theta(g_1g_2) = \theta(g_1)\theta(g_2) \quad \text{for all } g_1, g_2 \in G.$$

Notation:

$$G \cong H.$$

What does an isomorphism preserve?

An isomorphism preserves every property expressible only in terms of the group operation.

Examples of **isomorphism invariants**:

- the number of elements;
- Abelian versus non-Abelian;
- cyclic versus non-cyclic;
- orders of elements;
- number of solutions to equations such as $x^2 = e$;
- subgroup structure.

These invariants are useful for proving that two groups are **not** isomorphic.

Theorem 5.3.1: identity and inverses are preserved

Let $\theta : G \rightarrow H$ be an isomorphism.

Then

$$\theta(e_G) = e_H$$

and

$$\theta(g^{-1}) = \theta(g)^{-1}.$$

Proof idea. For every $g \in G$,

$$\theta(e_G)\theta(g) = \theta(e_G g) = \theta(g).$$

So $\theta(e_G)$ acts as the identity in H .

Similarly,

$$\theta(g)\theta(g^{-1}) = \theta(gg^{-1}) = \theta(e_G) = e_H.$$

Cyclic groups and the notation C_n

Any two cyclic groups with the same number of elements are isomorphic.

If

$$G = \langle g \rangle, \quad H = \langle h \rangle,$$

and both have order n , define

$$\theta(g^k) = h^k.$$

This is a well-defined isomorphism.

Therefore we write

$$C_n$$

for “the” cyclic group with n elements, up to isomorphism.

Additive notation often uses $\mathbb{Z}_n$ instead.

Direct products: definition

Let G and H be groups.

The **direct product** $G \times H$ has elements

$$(g, h), \quad g \in G, \quad h \in H.$$

The group operation is coordinatewise:

$$(g, h)(g', h') = (gg', hh').$$

Important point:

The product keeps elements of G and H in separate coordinates.

Theorem 5.3.2: direct products are groups

For any groups G and H , the direct product $G \times H$ is a group.

- Identity:

$$(e_G, e_H).$$

- Inverse:

$$(g, h)^{-1} = (g^{-1}, h^{-1}).$$

- Associativity follows coordinatewise from associativity in G and H .
- If G and H are finite, then

$$o(G \times H) = o(G)o(H).$$

Also $G \times H \cong H \times G$.

Examples of direct products

- $C_2 \times C_2$ has four elements.
- $C_4 \times C_2$ has eight elements.
- $S_3 \times S_3$ has $6 \cdot 6 = 36$ elements and is non-Abelian.
- If both factors are Abelian, the direct product is Abelian.

$([0]_2, [1]_2)$	$([1]_2, [1]_2)$
$([0]_2, [0]_2)$	$([1]_2, [0]_2)$

$\mathbb{Z}_2 \times \mathbb{Z}_2$ as a coordinate grid

C_4 versus the Klein four group

There are two different group structures of order 4.

Cyclic group C_4

$$C_4 = \{e, g, g^2, g^3\}$$

Element orders:

$$1, 4, 2, 4.$$

Thus $C_4 \not\cong C_2 \times C_2$.

Klein four group $C_2 \times C_2$

$$V_4 = \{e, a, b, c\}$$

with

$$a^2 = b^2 = c^2 = e.$$

All non-identity elements have order 2.